1.2.12 Registered RAM / Buffered RAM

This type of RAM module had additional storage areas - called buffers or registers - where the data is stored temporarily and checked for data integrity before being transferred. Similar in use to ECC RAM, though using a different method, this type of RAM is needed in servers, which require high levels of data integrity.

1.2.13 SD RAM

Synchronous Dynamic RAM was an improved version of DRAM that synchronised all its functions to a single frequency, usually the system's FSB frequency. The frequency refers to the rate at which the RAM would perform an action, namely refreshing, reading, or storing. SD RAM is available in various frequencies, with the fastest modules capable of running at 266 MHz (high performance modules, targeted at enthusiasts, which perform at higher frequencies, are also available). SD RAM modules have 168 contacts. After the release of DDR SDRAM, the original SDRAM began to be referred to as SDR SDRAM (Single Data Rate SDRAM).

1.3 Processor

1.3.1 32-bit / 64-bit CPU

"32/64-bit" refers to the width of the address bus and registers used by the CPU core. A 64-bit CPU has a 64-bit wide address bus and 64-bit wide registers. CPUs access data in the RAM by using the address bus. The breadth of the bus influences the amount of RAM that can be addressed. A 32-bit bus allows a maximum of 4 GB of RAM to be accessed. A 64-bit CPU, obviously, allows for more RAM to be used - about 16 exabytes (1 exabyte = 1,000,000,000 gigabytes, approximately). A 64-bit register can store 64 bits of data simultaneously. To fully utilise a 64-bit CPU, the operating system and application need to support the 64-bit mode of operation.

1.3.2 BGA

A Ball Grid Array (BGA) package is similar to a PGA package, except that the role of the pins is taken over by small balls of conductive